

Literature Status Note: ASQ Renorming and Dual-ASQ Questions

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Status. `literature_already_answered`. This note records two exact open-problem signals in Abrahamsen–Langemets–Lima, *Almost square Banach spaces*, arXiv:1402.0818, that are answered by later papers.

Source metadata. Trond A. Abrahamsen, Johann Langemets, and Vegard Lima, *Almost square Banach spaces*, arXiv:1402.0818. A local copy is stored as `source_paper.pdf`.

Question A: removing separability from the c_0 renorming theorem. In Section 2, immediately after Theorem 2.9 (label `thm:c0-asq`; source lines 526–561), the source proves that a separable Banach space can be equivalently renormed to be almost square (ASQ) if and only if it contains c_0 , and then asks whether the same conclusion holds for a general Banach space.

This is explicitly answered by Julio Becerra Guerrero, Gines Lopez-Perez, and Abraham Rueda Zoca, *Some results on almost square Banach spaces*, arXiv:1512.00610. Their introduction says that the main aim is to give a positive answer to the above question. Theorem 2.2 (label `renormageneral`) proves that every Banach space containing an isomorphic copy of c_0 admits an equivalent ASQ norm. Corollary 2.3 (label `carac0asq`) gives the exact characterization: a Banach space admits an equivalent ASQ norm if and only if it contains an isomorphic copy of c_0 . A local copy is stored as `supporting_paper_1512.00610.pdf`.

Question B: can a dual space be ASQ? In Section 6, Question 6.6 (source lines 1438–1442), the source asks whether the dual X^* of a Banach space X can be ASQ.

This is explicitly answered by Trond A. Abrahamsen, Petr Hajek, and Stanimir Troyanski, *Almost square dual Banach spaces*, arXiv:1912.12519. Their introduction cites Question 6.6 from arXiv:1402.0818 and says that they give a positive answer to whether ASQ spaces can be dual. Theorem 3.4 (label `thm:bidual-ell-infty-asq`) constructs a bidual ASQ renorming of ℓ_∞ . Theorem 3.7 (label `thm:dual-c0-asq-renorming`) proves the stronger characterization that a dual Banach space admits a dual ASQ norm if and only if it contains an isomorphic copy of c_0 . A local copy is stored as `supporting_paper_1912.12519.pdf`.

Scope limitation. This packet is a literature-status packet only. It should not be counted as a new solution, partial theorem, or counterexample. It covers the two ASQ renorming/dual-ASQ signals above. The separate LASQ versus WASQ separation question in arXiv:1402.0818 is not classified by this packet.

References

- [1] T. A. Abrahamsen, J. Langemets, and V. Lima, *Almost square Banach spaces*, arXiv:1402.0818.
- [2] J. Becerra Guerrero, G. Lopez-Perez, and A. Rueda Zoca, *Some results on almost square Banach spaces*, arXiv:1512.00610.
- [3] T. A. Abrahamsen, P. Hajek, and S. Troyanski, *Almost square dual Banach spaces*, arXiv:1912.12519.